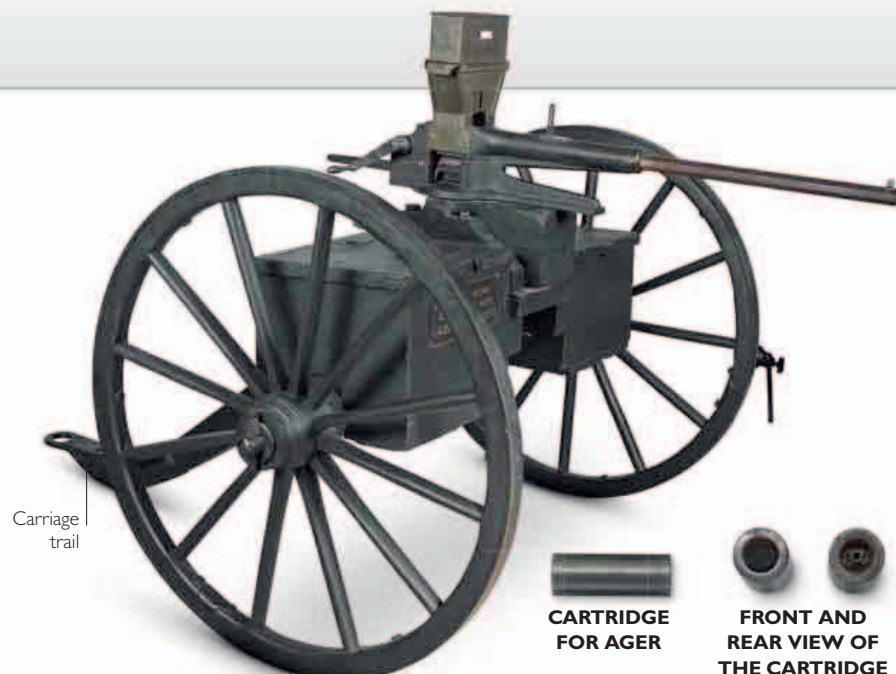


EARLY MACHINE-GUNS

By the time of the American Civil War (1861–65), there was widespread military interest in the potential benefit offered by rapid-fire weapons during combat. Two designers in particular, Wilson Ager and Richard Gatling, developed guns which offered considerable potential. Ager and Gatling's early "machine-guns" used a primitive type of cartridge in the form of reloadable steel tubes fitted with percussion caps, and consequently suffered from ammunition problems. However, the development of reliable unitary, metallic-cased center-fire cartridges (see pp.112–13), carrying propellant, projectile, and primer in one package, enabled these guns, and a number of other effective hand-cranked repeating guns, to achieve high rates of fire.



▲ AGER MACHINE-GUN

Date c.1860

Origin US

Length (Barrel) 3ft (.88m)

Caliber .58in

This gun was developed by Wilson Ager, and advertised by him as "an army in six square feet" because of its ability to fire 120 rounds per minute. Sixty guns were ordered for the Union Army, but barrel overheating problems meant the guns saw little use.

Ammunition hopper
(metal box on top of the
gun containing cartridges)



▲ EARLY GATLING CONVERTED TO METALLIC CARTRIDGE

Date c.1862

Origin US

Length (Barrel) 4½ft (1.4m)

Caliber .50in

Richard Jordan Gatling patented his hand-cranked, multi-barreled gun in 1862, and first used reloadable steel cartridges fired by percussion caps. Problems with misfires were common. In order to solve these issues, this early machine-gun was eventually modified to utilize the improved unitary cartridges.

